

GED Mathematics Worksheet - Rational Expressions and Functions – Full Solutions

Part 1: Multiplying Rational Expressions (Questions 1-10)

Question 1

Problem: Multiply: $\frac{3x}{4} \cdot \frac{8}{9x}$

Solution:

$$\begin{aligned}\frac{3x}{4} \cdot \frac{8}{9x} &= \frac{3x \cdot 8}{4 \cdot 9x} \\ &= \frac{24x}{36x} \\ &= \frac{24}{36} \quad (\text{cancel } x) \\ &= \frac{2}{3} \quad (\text{simplify by dividing by 12})\end{aligned}$$

Answer: $\boxed{\frac{2}{3}}$

Question 2

Problem: Multiply: $\frac{5a^2}{6b} \cdot \frac{12b}{10a}$

Solution:

$$\begin{aligned}\frac{5a^2}{6b} \cdot \frac{12b}{10a} &= \frac{5a^2 \cdot 12b}{6b \cdot 10a} \\ &= \frac{60a^2b}{60ab} \\ &= \frac{a^2}{a} \quad (\text{cancel } 60b) \\ &= a\end{aligned}$$

Answer: $\boxed{a}$

Question 3

Problem: Multiply: $\frac{x+2}{x-1} \cdot \frac{x-1}{x+3}$

Solution:

$$\begin{aligned}\frac{x+2}{x-1} \cdot \frac{x-1}{x+3} &= \frac{(x+2)(x-1)}{(x-1)(x+3)} \\ &= \frac{x+2}{x+3} \quad (\text{cancel } (x-1))\end{aligned}$$

Restrictions: $x \neq 1, -3$

Answer: $\boxed{\frac{x+2}{x+3}}$

Question 4

Problem: Multiply: $\frac{2x^2}{5y} \cdot \frac{15y^2}{8x}$

Solution:

$$\begin{aligned}\frac{2x^2}{5y} \cdot \frac{15y^2}{8x} &= \frac{2x^2 \cdot 15y^2}{5y \cdot 8x} \\ &= \frac{30x^2y^2}{40xy} \\ &= \frac{30xy}{40} \quad (\text{cancel } x \text{ and } y) \\ &= \frac{3xy}{4} \quad (\text{divide by } 10)\end{aligned}$$

Answer: $\boxed{\frac{3xy}{4}}$

Question 5

Problem: Multiply: $\frac{(x+4)(x-3)}{(x+1)} \cdot \frac{(x+5)}{(x-3)}$

Solution:

$$\begin{aligned}\frac{(x+4)(x-3)}{(x+1)} \cdot \frac{(x+5)}{(x-3)} &= \frac{(x+4)(x-3)(x+5)}{(x+1)(x-3)} \\ &= \frac{(x+4)(x+5)}{(x+1)} \quad (\text{cancel } (x-3)) \\ &= \frac{x^2 + 5x + 4x + 20}{x+1} \quad (\text{FOIL}) \\ &= \frac{x^2 + 9x + 20}{x+1}\end{aligned}$$

Restrictions: $x \neq -1, 3$

Answer: $\boxed{\frac{(x+4)(x+5)}{x+1} \text{ or } \frac{x^2 + 9x + 20}{x+1}}$

Question 6

Problem: Multiply: $\frac{x^2-4}{x^2+5x+6} \cdot \frac{x+3}{x-2}$

Solution:

First, factor all numerators and denominators:

$$\begin{aligned}x^2 - 4 &= (x - 2)(x + 2) \\x^2 + 5x + 6 &= (x + 2)(x + 3)\end{aligned}$$

Now multiply:

$$\begin{aligned}\frac{(x - 2)(x + 2)}{(x + 2)(x + 3)} \cdot \frac{x + 3}{x - 2} &= \frac{(x - 2)(x + 2)(x + 3)}{(x + 2)(x + 3)(x - 2)} \\&= 1 \quad (\text{cancel all factors})\end{aligned}$$

Restrictions: $x \neq -3, -2, 2$

Answer: $\boxed{1}$

Question 7

Problem: Multiply: $\frac{6m^2n}{7} \cdot \frac{14}{18mn^2}$

Solution:

$$\begin{aligned}\frac{6m^2n}{7} \cdot \frac{14}{18mn^2} &= \frac{6m^2n \cdot 14}{7 \cdot 18mn^2} \\&= \frac{84m^2n}{126mn^2} \\&= \frac{84m}{126n} \quad (\text{cancel } n) \\&= \frac{4m}{6n} \quad (\text{divide by 21}) \\&= \frac{2m}{3n} \quad (\text{divide by 2})\end{aligned}$$

Answer: $\boxed{\frac{2m}{3n}}$

Question 8

Problem: Multiply: $\frac{x^2-9}{x^2-1} \cdot \frac{x+1}{x+3}$

Solution:

Factor:

$$\begin{aligned}x^2 - 9 &= (x - 3)(x + 3) \\x^2 - 1 &= (x - 1)(x + 1)\end{aligned}$$

Multiply:

$$\begin{aligned}\frac{(x-3)(x+3)}{(x-1)(x+1)} \cdot \frac{x+1}{x+3} &= \frac{(x-3)(x+3)(x+1)}{(x-1)(x+1)(x+3)} \\ &= \frac{x-3}{x-1} \quad (\text{cancel } (x+3) \text{ and } (x+1))\end{aligned}$$

Restrictions: $x \neq 1, -1, -3$

Answer: $\boxed{\frac{x-3}{x-1}}$

Question 9

Problem: Multiply: $\frac{a^2-a-6}{a^2+4a+3} \cdot \frac{a+1}{a-3}$

Solution:

Factor:

$$\begin{aligned}a^2 - a - 6 &= (a-3)(a+2) \quad (\text{factors of } -6 \text{ that differ by } -1: -3, 2) \\ a^2 + 4a + 3 &= (a+1)(a+3) \quad (\text{factors of } 3 \text{ that sum to } 4: 1, 3)\end{aligned}$$

Multiply:

$$\begin{aligned}\frac{(a-3)(a+2)}{(a+1)(a+3)} \cdot \frac{a+1}{a-3} &= \frac{(a-3)(a+2)(a+1)}{(a+1)(a+3)(a-3)} \\ &= \frac{a+2}{a+3} \quad (\text{cancel } (a-3) \text{ and } (a+1))\end{aligned}$$

Restrictions: $a \neq -1, -3, 3$

Answer: $\boxed{\frac{a+2}{a+3}}$

Question 10

Problem: Multiply: $\frac{2x^2-8x}{x^2-16} \cdot \frac{x+4}{2x}$

Solution:

Factor:

$$\begin{aligned}2x^2 - 8x &= 2x(x-4) \\ x^2 - 16 &= (x-4)(x+4)\end{aligned}$$

Multiply:

$$\begin{aligned}\frac{2x(x-4)}{(x-4)(x+4)} \cdot \frac{x+4}{2x} &= \frac{2x(x-4)(x+4)}{(x-4)(x+4)(2x)} \\ &= 1 \quad (\text{cancel all factors})\end{aligned}$$

Restrictions: $x \neq 0, 4, -4$

Answer: $\boxed{1}$

Part 2: Dividing Rational Expressions (Questions 11-20)

Question 11

Problem: Divide: $\frac{5}{8} \div \frac{3}{4}$

Solution:

$$\begin{aligned}\frac{5}{8} \div \frac{3}{4} &= \frac{5}{8} \cdot \frac{4}{3} \quad (\text{invert and multiply}) \\ &= \frac{5 \cdot 4}{8 \cdot 3} \\ &= \frac{20}{24} \\ &= \frac{5}{6} \quad (\text{divide by 4})\end{aligned}$$

Answer: $\boxed{\frac{5}{6}}$

Question 12

Problem: Divide: $\frac{x^2}{y} \div \frac{x}{2y}$

Solution:

$$\begin{aligned}\frac{x^2}{y} \div \frac{x}{2y} &= \frac{x^2}{y} \cdot \frac{2y}{x} \quad (\text{invert and multiply}) \\ &= \frac{x^2 \cdot 2y}{y \cdot x} \\ &= \frac{2x^2y}{xy} \\ &= 2x \quad (\text{cancel } xy)\end{aligned}$$

Answer: $\boxed{2x}$

Question 13

Problem: Divide: $\frac{3a}{5b} \div \frac{9a^2}{10b^2}$

Solution:

$$\begin{aligned}\frac{3a}{5b} \div \frac{9a^2}{10b^2} &= \frac{3a}{5b} \cdot \frac{10b^2}{9a^2} \quad (\text{invert and multiply}) \\ &= \frac{3a \cdot 10b^2}{5b \cdot 9a^2} \\ &= \frac{30ab^2}{45a^2b} \\ &= \frac{30b}{45a} \quad (\text{cancel } ab) \\ &= \frac{2b}{3a} \quad (\text{divide by 15})\end{aligned}$$

Answer: $\boxed{\frac{2b}{3a}}$

Question 14

Problem: Divide: $\frac{m^2n}{6} \div \frac{mn^2}{12}$

Solution:

$$\begin{aligned}\frac{m^2n}{6} \div \frac{mn^2}{12} &= \frac{m^2n}{6} \cdot \frac{12}{mn^2} \quad (\text{invert and multiply}) \\ &= \frac{m^2n \cdot 12}{6 \cdot mn^2} \\ &= \frac{12m^2n}{6mn^2} \\ &= \frac{12m}{6n} \quad (\text{cancel } mn) \\ &= 2m \quad (\text{divide by 6})\end{aligned}$$

Answer: $\boxed{2m}$

Question 15

Problem: Divide: $\frac{x+2}{x-3} \div \frac{x+2}{x+5}$

Solution:

$$\begin{aligned}\frac{x+2}{x-3} \div \frac{x+2}{x+5} &= \frac{x+2}{x-3} \cdot \frac{x+5}{x+2} \quad (\text{invert and multiply}) \\ &= \frac{(x+2)(x+5)}{(x-3)(x+2)} \\ &= \frac{x+5}{x-3} \quad (\text{cancel } (x+2))\end{aligned}$$

Restrictions: $x \neq -2, 3, -5$

Answer: $\boxed{\frac{x+5}{x-3}}$

Question 16

Problem: Divide: $\frac{x^2-4}{x^2+6x+9} \div \frac{x-2}{x+3}$

Solution:

Factor:

$$\begin{aligned}x^2 - 4 &= (x - 2)(x + 2) \\x^2 + 6x + 9 &= (x + 3)^2\end{aligned}$$

Divide:

$$\begin{aligned}\frac{(x-2)(x+2)}{(x+3)^2} \div \frac{x-2}{x+3} &= \frac{(x-2)(x+2)}{(x+3)^2} \cdot \frac{x+3}{x-2} \quad (\text{invert and multiply}) \\&= \frac{(x-2)(x+2)(x+3)}{(x+3)^2(x-2)} \\&= \frac{x+2}{x+3} \quad (\text{cancel } (x-2) \text{ and one } (x+3))\end{aligned}$$

Restrictions: $x \neq -3, 2$

Answer: $\boxed{\frac{x+2}{x+3}}$

Question 17

Problem: Divide: $\frac{x^2-8x}{x^2+8x+16} \div \frac{x-8}{x^2-2x-24}$

Solution:

Factor:

$$\begin{aligned}x^2 - 8x &= x(x - 8) \\x^2 + 8x + 16 &= (x + 4)^2 \\x^2 - 2x - 24 &= (x - 6)(x + 4) \quad (\text{factors of } -24 \text{ that differ by } -2: -6, 4)\end{aligned}$$

Divide:

$$\begin{aligned}\frac{x(x-8)}{(x+4)^2} \div \frac{x-8}{(x-6)(x+4)} &= \frac{x(x-8)}{(x+4)^2} \cdot \frac{(x-6)(x+4)}{x-8} \quad (\text{invert and multiply}) \\&= \frac{x(x-8)(x-6)(x+4)}{(x+4)^2(x-8)} \\&= \frac{x(x-6)}{x+4} \quad (\text{cancel } (x-8) \text{ and one } (x+4))\end{aligned}$$

Restrictions: $x \neq 0, -4, 8, 6$

Answer: $\boxed{\frac{x(x-6)}{x+4}}$

Question 18

Problem: Divide: $\frac{2x^2+5x+3}{x^2-1} \div \frac{2x+3}{x-1}$

Solution:

Factor:

$$2x^2 + 5x + 3 = (2x + 3)(x + 1) \quad (\text{factors of 6 that sum to 5: } 2, 3)$$

$$x^2 - 1 = (x - 1)(x + 1)$$

Divide:

$$\begin{aligned} \frac{(2x+3)(x+1)}{(x-1)(x+1)} \div \frac{2x+3}{x-1} &= \frac{(2x+3)(x+1)}{(x-1)(x+1)} \cdot \frac{x-1}{2x+3} \quad (\text{invert and multiply}) \\ &= \frac{(2x+3)(x+1)(x-1)}{(x-1)(x+1)(2x+3)} \\ &= 1 \quad (\text{cancel all factors}) \end{aligned}$$

Restrictions: $x \neq 1, -1, -\frac{3}{2}$

Answer: $\boxed{1}$

Question 19

Problem: Divide: $\frac{a^2-9}{a^2-4} \div \frac{a+3}{a-2}$

Solution:

Factor:

$$a^2 - 9 = (a - 3)(a + 3)$$

$$a^2 - 4 = (a - 2)(a + 2)$$

Divide:

$$\begin{aligned} \frac{(a-3)(a+3)}{(a-2)(a+2)} \div \frac{a+3}{a-2} &= \frac{(a-3)(a+3)}{(a-2)(a+2)} \cdot \frac{a-2}{a+3} \quad (\text{invert and multiply}) \\ &= \frac{(a-3)(a+3)(a-2)}{(a-2)(a+2)(a+3)} \\ &= \frac{a-3}{a+2} \quad (\text{cancel } (a+3) \text{ and } (a-2)) \end{aligned}$$

Restrictions: $a \neq 2, -2, -3$

Answer: $\boxed{\frac{a-3}{a+2}}$

Question 20

Problem: Divide: $\frac{x^2-5x+6}{x^2+x-12} \div \frac{x-2}{x+4}$

Solution:

Factor:

$$x^2 - 5x + 6 = (x - 2)(x - 3) \quad (\text{factors of 6 that sum to } -5: -2, -3)$$

$$x^2 + x - 12 = (x + 4)(x - 3) \quad (\text{factors of } -12 \text{ that sum to } 1: 4, -3)$$

Divide:

$$\begin{aligned} \frac{(x-2)(x-3)}{(x+4)(x-3)} \div \frac{x-2}{x+4} &= \frac{(x-2)(x-3)}{(x+4)(x-3)} \cdot \frac{x+4}{x-2} && (\text{invert and multiply}) \\ &= \frac{(x-2)(x-3)(x+4)}{(x+4)(x-3)(x-2)} \\ &= 1 && (\text{cancel all factors}) \end{aligned}$$

Restrictions: $x \neq -4, 3, 2$

Answer: $\boxed{1}$

Part 3: Understanding Functions (Questions 21-30)

Question 21

Problem: Determine if the relation $\{(1, 2), (2, 4), (3, 6), (4, 8)\}$ is a function. Explain why or why not.

Solution:

A relation is a function if each element in the domain is paired with exactly one element in the range.

Domain: $\{1, 2, 3, 4\}$ Range: $\{2, 4, 6, 8\}$

Checking each domain element:

- $1 \rightarrow 2$ (one output)
- $2 \rightarrow 4$ (one output)
- $3 \rightarrow 6$ (one output)
- $4 \rightarrow 8$ (one output)

Each domain element is paired with exactly one range element.

Answer: Yes, it is a function.

Question 22

Problem: Determine if the relation $\{(1, 3), (2, 5), (1, 7), (3, 9)\}$ is a function. Explain why or why not.

Solution:

Domain: $\{1, 2, 3\}$ Range: $\{3, 5, 7, 9\}$

Checking each domain element:

- $1 \rightarrow 3$ and $1 \rightarrow 7$ (two outputs!)
- $2 \rightarrow 5$ (one output)
- $3 \rightarrow 9$ (one output)

The domain element 1 is paired with two different range elements (3 and 7).

Answer: No, it is not a function.

Question 23

Problem: For $f(x) = 3x - 2$, find $f(5)$.

Solution:

Substitute $x = 5$ into the function:

$$\begin{aligned} f(5) &= 3(5) - 2 \\ &= 15 - 2 \\ &= 13 \end{aligned}$$

Answer: $f(5) = 13$

Question 24

Problem: For $g(x) = x^2 + 4$, find $g(-3)$.

Solution:

Substitute $x = -3$ into the function:

$$\begin{aligned} g(-3) &= (-3)^2 + 4 \\ &= 9 + 4 \\ &= 13 \end{aligned}$$

Answer: $g(-3) = 13$

Question 25

Problem: For $h(x) = 2x^2 - 5x + 1$, find $h(2)$.

Solution:

Substitute $x = 2$ into the function:

$$\begin{aligned} h(2) &= 2(2)^2 - 5(2) + 1 \\ &= 2(4) - 10 + 1 \\ &= 8 - 10 + 1 \\ &= -1 \end{aligned}$$

Answer: $h(2) = -1$

Question 26

Problem: For $f(x) = \frac{x+3}{x-1}$, find $f(4)$.

Solution:

Substitute $x = 4$ into the function:

$$\begin{aligned} f(4) &= \frac{4+3}{4-1} \\ &= \frac{7}{3} \end{aligned}$$

Answer: $f(4) = \frac{7}{3}$

Question 27

Problem: For $g(x) = 5x - 7$, find $g(-2)$.

Solution:

Substitute $x = -2$ into the function:

$$\begin{aligned}g(-2) &= 5(-2) - 7 \\&= -10 - 7 \\&= -17\end{aligned}$$

Answer: $g(-2) = -17$

Question 28

Problem: For $h(x) = x^3 - 2x + 4$, find $h(3)$.

Solution:

Substitute $x = 3$ into the function:

$$\begin{aligned}h(3) &= (3)^3 - 2(3) + 4 \\&= 27 - 6 + 4 \\&= 25\end{aligned}$$

Answer: $h(3) = 25$

Question 29

Problem: For $f(x) = \sqrt{x+5}$, find $f(4)$.

Solution:

Substitute $x = 4$ into the function:

$$\begin{aligned}f(4) &= \sqrt{4+5} \\&= \sqrt{9} \\&= 3\end{aligned}$$

Answer: $f(4) = 3$

Question 30

Problem: For $g(x) = |2x - 3|$, find $g(1)$.

Solution:

Substitute $x = 1$ into the function:

$$\begin{aligned}g(1) &= |2(1) - 3| \\&= |2 - 3| \\&= |-1| \\&= 1\end{aligned}$$

Answer: $g(1) = 1$

Part 4: Evaluating Functions with Expressions (Questions 31-35)

Question 31

Problem: For $g(x) = 3x + 5$, find $g(4n)$.

Solution:

Substitute $x = 4n$ into the function:

$$\begin{aligned} g(4n) &= 3(4n) + 5 \\ &= 12n + 5 \end{aligned}$$

Answer: $g(4n) = 12n + 5$

Question 32

Problem: For $h(x) = 2x^2 - 3x + 7$, find $h(4y^2)$.

Solution:

Substitute $x = 4y^2$ into the function:

$$\begin{aligned} h(4y^2) &= 2(4y^2)^2 - 3(4y^2) + 7 \\ &= 2(16y^4) - 12y^2 + 7 \\ &= 32y^4 - 12y^2 + 7 \end{aligned}$$

Answer: $h(4y^2) = 32y^4 - 12y^2 + 7$

Question 33

Problem: For $f(x) = x^2 - 6x$, find $f(a + 2)$.

Solution:

Substitute $x = a + 2$ into the function:

$$\begin{aligned} f(a + 2) &= (a + 2)^2 - 6(a + 2) \\ &= a^2 + 4a + 4 - 6a - 12 \\ &= a^2 - 2a - 8 \end{aligned}$$

Answer: $f(a + 2) = a^2 - 2a - 8$

Question 34

Problem: For $g(x) = 5x - 1$, find $g(2m + 3)$.

Solution:

Substitute $x = 2m + 3$ into the function:

$$\begin{aligned} g(2m + 3) &= 5(2m + 3) - 1 \\ &= 10m + 15 - 1 \\ &= 10m + 14 \end{aligned}$$

Answer: $g(2m + 3) = 10m + 14$

Question 35

Problem: For $h(x) = x^2 + 2x - 8$, find $h(x - 1)$.

Solution:

Substitute $x = x - 1$ into the function:

$$\begin{aligned} h(x - 1) &= (x - 1)^2 + 2(x - 1) - 8 \\ &= x^2 - 2x + 1 + 2x - 2 - 8 \\ &= x^2 - 9 \end{aligned}$$

Wait, let me recalculate more carefully:

$$\begin{aligned} h(x - 1) &= (x - 1)^2 + 2(x - 1) - 8 \\ &= (x^2 - 2x + 1) + (2x - 2) - 8 \\ &= x^2 - 2x + 1 + 2x - 2 - 8 \\ &= x^2 + (-2x + 2x) + (1 - 2 - 8) \\ &= x^2 - 9 \end{aligned}$$

Actually, let me verify once more by expanding step-by-step:

$$\begin{aligned} h(x - 1) &= (x - 1)^2 + 2(x - 1) - 8 \\ &= x^2 - 2x + 1 + 2x - 2 - 8 \\ &= x^2 + (-2x + 2x) + (1 - 2 - 8) \\ &= x^2 + 0 + (-9) \\ &= x^2 - 9 \end{aligned}$$

Hmm, but this doesn't match the answer key. Let me reconsider. The original function is $h(x) = x^2 + 2x - 8$. When we substitute $x - 1$:

$$\begin{aligned} h(x - 1) &= (x - 1)^2 + 2(x - 1) - 8 \\ &= x^2 - 2x + 1 + 2x - 2 - 8 \\ &= x^2 - 9 \end{aligned}$$

Wait, I think the answer key might have a different interpretation. Let me check if they meant to find $h(t)$ where $t = x - 1$, then express in terms of x . Actually, the standard interpretation is what I did. But the answer key says $x^2 - 4x - 5$. Let me recalculate:

Actually, I realize my error. Let me be more careful:

$$h(x-1) = (x-1)^2 + 2(x-1) - 8$$

Let $u = x - 1$. Then:

$$h(u) = u^2 + 2u - 8$$

So:

$$\begin{aligned} h(x-1) &= (x-1)^2 + 2(x-1) - 8 \\ &= x^2 - 2x + 1 + 2x - 2 - 8 \\ &= x^2 - 9 \end{aligned}$$

But the answer key says $x^2 - 4x - 5$. Let me check if there's a different function. Actually, I think there might be an error in my answer key. Let me verify by a different method. If $h(x) = x^2 + 2x - 8$, then:

$$h(x-1) = (x-1)^2 + 2(x-1) - 8 = x^2 - 2x + 1 + 2x - 2 - 8 = x^2 - 9$$

This is correct. However, the provided answer key says $x^2 - 4x - 5$. Let me check if perhaps the original function was different. Looking back at the worksheet, it says $h(x) = x^2 + 2x - 8$.

Actually, I wonder if the answer key had a typo. Let me just provide the correct solution:

Answer: $h(x-1) = x^2 - 9$

Actually, wait. Let me reconsider the answer key. Perhaps I made an error. Let me expand very carefully:

$$h(x-1) = (x-1)^2 + 2(x-1) - 8$$

$$(x-1)^2 = x^2 - 2x + 1 \quad 2(x-1) = 2x - 2$$

$$\text{So: } h(x-1) = x^2 - 2x + 1 + 2x - 2 - 8 = x^2 + (-2x + 2x) + (1 - 2 - 8) = x^2 - 9$$

I'm confident this is correct. The answer key might have had a different problem in mind. Let me provide the correct answer:

Answer: $h(x-1) = x^2 - 9$

Part 5: Functions and Graphs (Questions 36-40)

Question 36

Problem: Use the vertical line test to determine if the graph of $y = x^2$ represents a function.

Solution:

The vertical line test states: A relation is a function if every vertical line intersects the graph at most once.

For $y = x^2$ (a parabola opening upward):

- The parabola is symmetric about the y -axis
- For any value of x , there is exactly one corresponding y -value
- Any vertical line drawn on the graph will intersect the parabola at exactly one point

Therefore, the graph passes the vertical line test.

Answer: Yes, $y = x^2$ represents a function.

Question 37

Problem: Use the vertical line test to determine if the graph of $x = y^2$ represents a function.

Solution:

For $x = y^2$ (a parabola opening to the right):

- This can be rewritten as $y = \pm\sqrt{x}$
- For any positive value of x , there are two corresponding y -values (one positive and one negative)
- For example, when $x = 4$: $y = 2$ or $y = -2$
- A vertical line at $x = 4$ would intersect the parabola at two points: $(4, 2)$ and $(4, -2)$

Therefore, the graph fails the vertical line test.

Answer: No, $x = y^2$ does not represent a function.

Question 38

Problem: For the function $f(x) = 2x - 3$, find $f(2)$ using the equation. Then verify your answer by substituting into the function.

Solution:

Using the equation:

$$\begin{aligned} f(2) &= 2(2) - 3 \\ &= 4 - 3 \\ &= 1 \end{aligned}$$

Verification by substitution:

$$\begin{aligned}f(x) &= 2x - 3 \\f(2) &= 2(2) - 3 = 4 - 3 = 1 \quad \checkmark\end{aligned}$$

Both methods give the same result.

Answer: $f(2) = 1$

Question 39

Problem: Describe the behavior of a function during the interval $-2 \leq x < 1$ if the function increases throughout this interval.

Solution:

If a function increases throughout the interval $-2 \leq x < 1$, this means:

- As x increases from -2 to values approaching 1, the y -values (function outputs) also increase
- The function has a positive slope or positive rate of change in this interval
- If we pick any two points x_1 and x_2 where $-2 \leq x_1 < x_2 < 1$, then $f(x_1) < f(x_2)$
- Graphically, the function rises from left to right throughout this interval
- The function reaches its minimum value at $x = -2$ and approaches its maximum value as x approaches 1 (but doesn't reach it since the interval is open at $x = 1$)

Answer: $\boxed{\text{The function increases as } x \text{ increases; the graph rises from left to right.}}$

Question 40

Problem: For a piecewise function, explain why different intervals might have different behaviors (increasing, decreasing, or constant).

Solution:

A piecewise function is defined by different rules (equations) on different intervals of the domain. Because each interval uses a different rule, the behavior can vary:

- **Different Rules:** Each piece of the function follows a different equation. For example:

$$f(x) = \begin{cases} x + 2 & \text{if } x < 0 \\ x^2 & \text{if } 0 \leq x < 3 \\ 5 & \text{if } x \geq 3 \end{cases}$$

- **Linear vs. Nonlinear:** One piece might be linear (constant slope), while another is quadratic (changing slope)

- **Different Slopes:** Linear pieces can have different slopes. For example, $f(x) = x + 2$ has slope 1, while $f(x) = -2x$ has slope -2
- **Behavior Examples:**
 - For $x < 0$: $f(x) = x + 2$ increases (positive slope)
 - For $0 \leq x < 3$: $f(x) = x^2$ increases (parabola opening upward)
 - For $x \geq 3$: $f(x) = 5$ is constant (horizontal line)
- **Transitions:** At the boundaries between intervals, the function can have jumps, corners, or smooth transitions depending on how the pieces connect

Answer: Different intervals use different rules, so each piece can have its own behavior.